Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Python · Django · Healthcare SaaS · MySQL · AWS · CI/CD · Production Reliability · REST APIs · Monitoring · Agile

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PROFESSIONAL SUMMARY

Software Engineer with 8 years of production Python experience building and maintaining reliable, high-quality SaaS applications - with a track record of owning production stability, systematic debugging, and CI/CD-driven deployment quality. At Entrata, maintained 99.5% uptime across 100K+ monthly active users, owned end-to-end QA for 20+ full-stack modules, and built monitoring and observability tooling that reduced triage time 40% across multi-tenant production environments. Built AI on FHIR - a HIPAA-compliant NLP backend for healthcare records - demonstrating genuine interest in software that improves healthcare outcomes. Comfortable across Python web frameworks (FastAPI, Flask), MySQL/PostgreSQL, AWS S3/EC2/RDS, REST APIs, and GitHub-based CI/CD workflows. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Engineering & Backend: Python, SQL, FastAPI, Flask, PHP, JavaScript, TypeScript, React, Bash, Docker, MySQL, PostgreSQL, Oracle, Django ORM patterns, AWS (S3, EC2, RDS), Redis, RabbitMQ, REST APIs, CI/CD (GitHub Actions, GitHub PR workflows), Monitoring & Logging, Linux, Git

Quality & Reliability: End-to-End QA Ownership, Test-Driven Development, Automated Testing, Bug Diagnosis & Root Cause Analysis, Production Debugging, Systematic Debugging, Code Review, Performance Optimization, Agile/Scrum, Azure DevOps-compatible workflows

AI & Data: LangGraph, LangChain, RAG Pipelines, NLP (BERT, spaCy), LLM Orchestration, ChromaDB, Model Evaluation, PyTorch, scikit-learn, PySpark, Tableau

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built Python agentic backend with end-to-end QA ownership - validated all outputs against compliance constraints before deployment; CI/CD-integrated automated testing enforcing reliability at every release.
- Diagnosed and resolved complex issues in distributed RAG pipeline; monitoring on retrieval quality and response groundedness with systematic debugging across edge cases.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built Python and R ML pipelines on WHO and UK Road Safety datasets - $R^2 = 0.977$; systematic debugging and validation across 190+ countries of data; end-to-end QA from ingestion through model output.
- Mentoring 600+ students in Python, SQL, and database design; designed monitoring dashboards improving research comprehension by 40%.

Entrata - PropTech / Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Owned production stability and reliability across 20+ SaaS modules** - 99.5% uptime at 100K+ monthly active users; resolved complex bugs and completed monthly upgrades across compliance-critical workflows; zero compliance violations across 7 years; systematic debugging and RCA for production incidents.
- **End-to-end QA ownership and CI/CD discipline** - owned automated testing and QA for all work before peer code review; GitHub PR workflows with daily deployment cadence; test-driven development across backend modules; reduced bug backlog to sub-10 for 6+ consecutive months.
- **Monitoring, logging, and production observability** - built internal monitoring tooling reducing triage time 40%; performance monitoring across multi-tenant environments; alert response and service request handling at production scale.
- **Python backend and MySQL/PostgreSQL database work** - Python web framework development (FastAPI, Flask patterns); 200+ SQL query optimizations (30% performance gain); schema design; AWS S3, EC2, RDS; REST API development across regulated SaaS platform.
- Mentoring 5+ junior engineers; 300+ peer code reviews; Agile/Scrum across 120+ sprints.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built property management backend modules (PHP, Python, MySQL, PostgreSQL); data pipeline for 1.6M HUD/Tax Credit records with systematic validation; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AI on FHIR: NLP Healthcare Records Query System

Healthcare · Python

Python · Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

Python + React healthcare platform translating natural-language queries into FHIR API calls using BERT embeddings and LLMs - reduced retrieval time ~60%; HIPAA-compliant RBAC with full data isolation across structured medical records. Built to improve healthcare data access outcomes.

AgentProbe: Automated Testing & Production Monitoring Platform

Testing · CI/CD

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

Python automated testing and production monitoring platform: 15-endpoint FastAPI service, 5-metric quality engine, CI-integrated regression gate blocking failures before merge, systematic monitoring and logging across 1,240+ test cases - zero regressions shipped to production.

Leasel: Python Agentic Backend with HITL Quality Controls

Python · Reliability

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Python distributed backend with end-to-end QA ownership: 8-agent system with conditional routing, systematic debugging across edge cases, and HITL escalation when confidence thresholds not met. 93.6% precision, 87.9% F1 across 1,240 test cases.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017